DP Capability Performance Analysis

KCG 3000t Vessel (Pacific No.20, 3020) Fixed Heading vs. Optimal Heading Mode Comparison

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This report is provided as a reference document only. The report represents the result of mathematical modeling based on formulas and general vessel parameters. Its main objective is to provide an overall understanding of the vessel's ability to maintain position under various weather conditions for specified parameters. It should be noted that the calculation does not take into account all specific and individual vessel characteristics, and therefore may differ from the vessel's actual performance, as in reality its behavior is influenced by numerous factors that cannot be fully predicted or simulated.

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1 General information

1.1 Overview

This study presents a Dynamic Capability Analysis aimed at establishing a quantitative basis for assessing the ability of a vessel to maintain its position under environmental conditions approaching Sea State 5. The analysis focuses on the station-keeping performance of the KCG 3000t vessel, with the objective of evaluating position-keeping limitations under dynamically varying environmental conditions.

The assessment is conducted for both fixed-heading and optimal-heading Dynamic Positioning (DP) operational modes. All analyses are performed in accordance with the Level 3 site requirements defined in the DNVGL-ST-0111 standard.

Vessel particulars and the environmental test-case scenarios considered in this study are summarized in item 3.1. Time-domain simulations are carried out for two distinct environmental condition scenarios, with the vessel model accurately representing the actual vessel equipment and configuration.

The simulations are performed using the Nav-Vessel time-domain simulation framework coupled with the latest version of the Navis DP software, developed by Navis Engineering Oy. This integrated simulation approach enables a realistic and robust evaluation of the vessel's dynamic positioning capability under challenging environmental conditions.

1.2 Ship motion simulator

The CotMACS-JDP v.2.00 ship motion simulator is used for performance testing.

It is based on the 6-degree mathematical model of the vessel. This mathematical model includes models of hydrodynamics, thrusters, power system, navigational sensors, environmental disturbances.

The ship motion control algorithms use ship motion parameters: fore-aft and athwartships forces, yaw rate, heading and coordinates. Also, it is necessary to know the parameters of current: speed and angle relative to a ship.

However, not all parameters can be measured quite accurately and some of them cannot be measured at all, therefore it is necessary to estimate these parameters. Parameters are connected to each other via ship motion equations, and the mathematical model allows their estimation.

Ship motion parameters also depend on external disturbances (wave), which cannot be measured. Wave disturbance is considered by wave filters.

For this reason, measured data and data received from the mathematical model are used together and are processed by the Kalman filter algorithm. The Kalman filter uses a ship motion mathematical model for predicting measurements. Ship position data comes from several sensors (with different

characteristics). Measurements, processed by filter, are weighed in accordance with their noise levels and then used for updating of ship state, received from the mathematical model.

Kalman filter algorithm allows generation of an optimal estimation of ship motion parameters and parameters of current. Also, if measurements are absent, such algorithms allow a comparatively accurate prediction of ship motion parameters for some period.

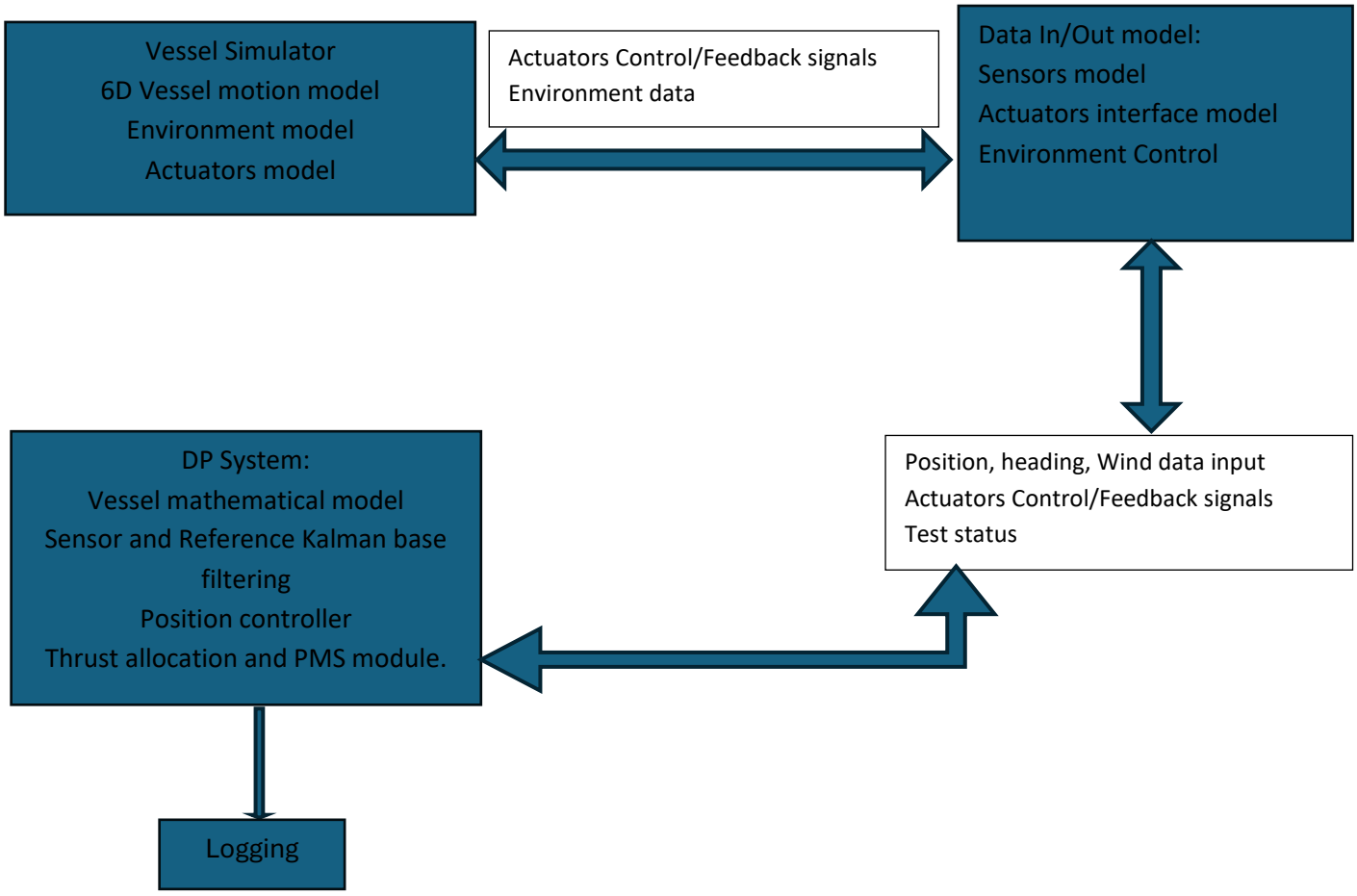
1.3 Vessel mathematical model

The mathematical model is a special calculating module (hydro- and aerodynamic ship description). This module continually calculates the ship's response from external disturbances as well as from thrusters, propulsions, and steering systems. The response is indirectly estimated from sensor data.

Wind speed and wind angles, relative to a ship, are parameters of wind disturbance. Wind forces and moments are calculated based on aerodynamic ship hull characteristics. Data from control systems sensors are parameters defining the propulsion thrust. Using these parameters, instantaneous forces from propulsion, rudders, and thrusters are calculated.

All ship particulars: mass, draft, dimensions, hull type, propulsion unit type, thruster types and structure, superstructure location, etc. are considered in the mathematical model, which is modified in accordance with sea trial data and then applicable to the actual ship.

The mathematical model adjustment is as accurate as possible and describes all possible modes of ship motion from external disturbances as well as thrusters, propulsions, and steering systems.

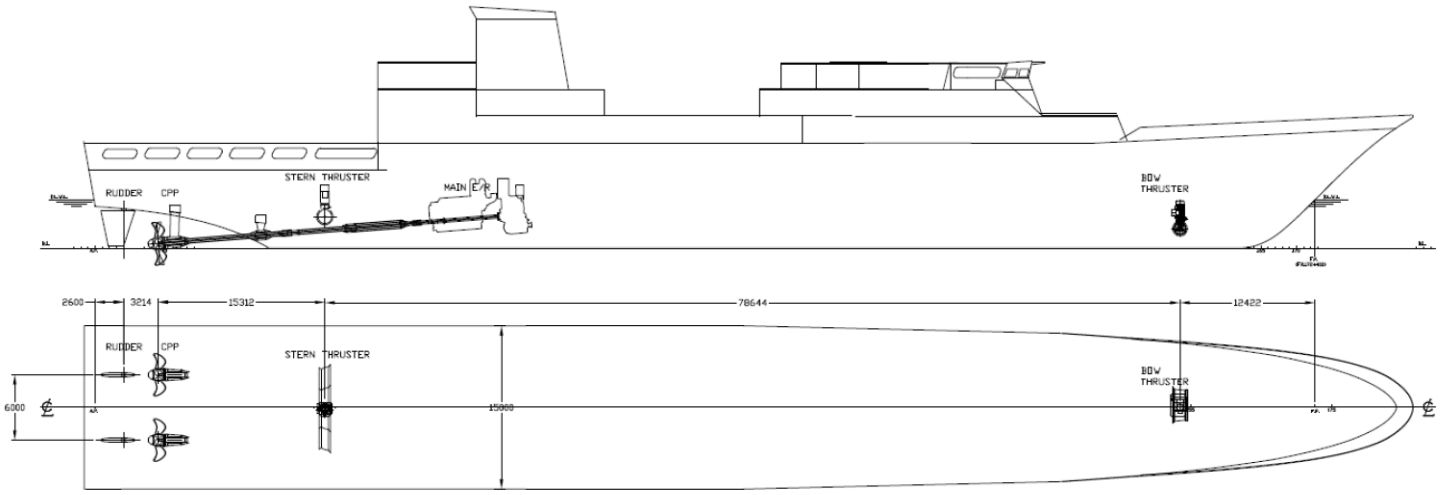


2 Vessel configuration

The following parameters are used in the simulator

1. **Vessel ID**: KCG 3000t (Pacific No.20, 3020)

2. **General arrangement and vessel data**



Hull Data		
Length over all	L_{OA}	122.3 m
Length between perpendiculars	L_{pp}	109.6 m
Length waterline	L_{WL}	112.3 m
Maximum breadth at water line	B	15.0 m
Draft (operating)		4.40 m
Depth (main deck->keel)		7.0 m
Longitudinal distance between the foremost and aftmost points on the hull below the free surface at summer load line draft even keel	L_{os}	112.3 m
Longitudinal position of $L_{os}/2$	x_{Los}	-1.3 m
Bow angle		16.9
Water plane area behind midship	A_{WLaft}	836 m^2
Transversal wind area	$A_{F,wind}$	200 m^2
Lateral wind area	$A_{L,wind}$	1067 m^2
Surge position of area centre of the projected longitudinal area above water with respect to $L_{pp}/2$	$x_{L,air}$	3.0 m
Projected longitudinal area below water	$A_{L,current}$	442 m^2
Surge position of area centre of the projected longitudinal area below water with respect to $L_{pp}/2$	$x_{L,current}$	1.8 m

3. Thruster and main propulsion data

Propulsor Details								
Thruster No & Thruster info	Power		Dia-meter	Thrust (ton)		Coeff.	Position	
	kW	hp	m	Max	Min	kg/hp	X, m	Y, m
#1 Bow Thruster (CPP)	350	469	1.05	5.4	0.0	11.6	42.4	0.0
#2 Stern Thruster (CPP)	450	603	1.20	6.7	0.0	11.1	-36.3	0.0
#3 Port Propul-sion (CPP)	1361	1825	3.94	20.7 (AH) 11.7 (AS)	0.0	11.3	-51.8	-3.0
#4 Stbd Propul-sion (CPP)	1361	1825	3.94	20.7 (AH) 11.7 (AS)	0.0	11.3	-51.8	3.0

4. Rudders

Rudders: two mechanically linked rudders

Max ruder angle: ±35˚

5. Power system parameters

Main Engine(1+1 DE): 1361kW x 500rpm

Bow thruster: 350kW

Stern thruster: 450kW

Power limitations: no limits set (up to 100% power is used during testing)

3 Test parameters

3.1 Weather conditions

The weather conditions are taken closer to the maximal edge of the Sea State 5

Current speed = 0.37 m/s
Wind Speed = 13.50 m/s
Wave height (significant, H1/3) = 4 m
Wave period: integrally calculation (~ 6sec)
according to Pierson-Moskowitz Spectrum (JONSWAP, gamma=1)

3.2 Test initial conditions

Test mode #1 (Fixed Heading):
DP mode: Auto position keeping mode + Auto heading keeping mode
Initial conditions:
Position X = 0,
Position Y = 0,
Heading = 0˚ (bow is oriented to the North)
Position reference system: DGNSS,
the sensor accuracy is considered less 0.5m

Test mode #2 (Optimal Heading):
DP mode: Auto position keeping mode + Optimal heading keeping mode
Initial conditions:
Position X = 0,
Position Y = 0,
Heading = 0˚ (bow is oriented to the North)

The Auto position keeping mode means that the system tries to keep the vessel's position at the preset point (X=0, Y=0).
The Auto heading mode means that the system tries to keep the vessel's heading at the preset value (0˚).
The Optimal heading keeping mode means that the system automatically adjusts the vessel's heading to keep the vessel's bow against the disturbing force (wind + current).

4 Test mode #1

The objective of this test is to determine the capability sectors of environmental disturbance directions within which the vessel is able to maintain both heading and position. The analysis identifies directional ranges of disturbing forces for which the vessel's station-keeping performance satisfies the defined acceptance criteria.

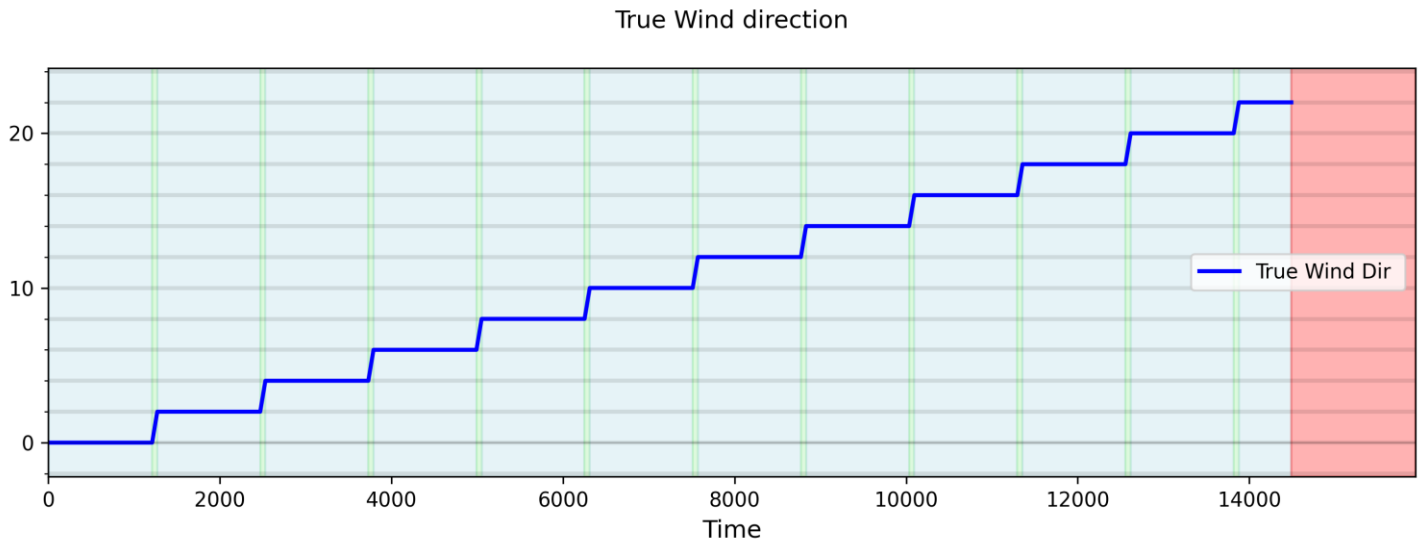
- vessel is set in conditions according to item 3.1 and 3.2 (Test mode #1).
- DP mode is active, and the vessel keeps the heading and position.
- initial time value for logging is set in 0 seconds.
- disturbing force change speed is 0.5deg/sec
- direction of the current is always collinear with the wind direction and they change simultaneously
- test interval is 2 degrees, the test duration is 20min
- test becomes failed if the vessel moves from the initial point more than 15m, after that the logging stops automatically
- test is conducted in a 0-180˚ range of disturbing force directions, because it is supposed that the results in a 180-360 ˚ part are identical due to the symmetric configuration and can be mirrored.

Test Execution Strategy:
Due to limited capability in beam seas (environmental forces from the side), testing was divided into two separate simulation runs:
1. ***Bow Sector Test:*** Environmental direction starting from 0° (bow-on) and rotating clockwise in 2° increments until the vessel's position-keeping capability limit was reached.
2. ***Stern Sector Test:*** Environmental direction starting from 180° (stern-on) and rotating counterclockwise in 2° increments until the vessel's position-keeping capability limit was reached.

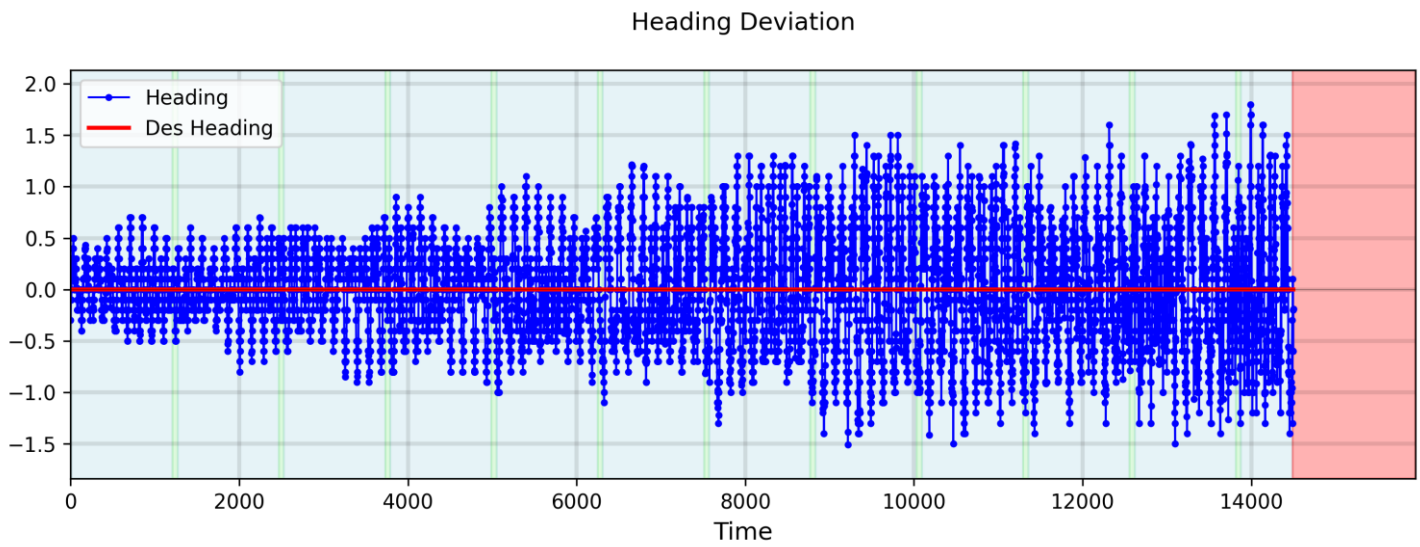
The angular range between the bow and stern capability limits represents environmental directions where the vessel is *unable to maintain position* under the specified environmental conditions (Sea State 5). These directions correspond to beam seas and quartering seas where lateral forces exceed the vessel's thruster capacity in Fixed Heading mode.

4.1 Test mode #1 results

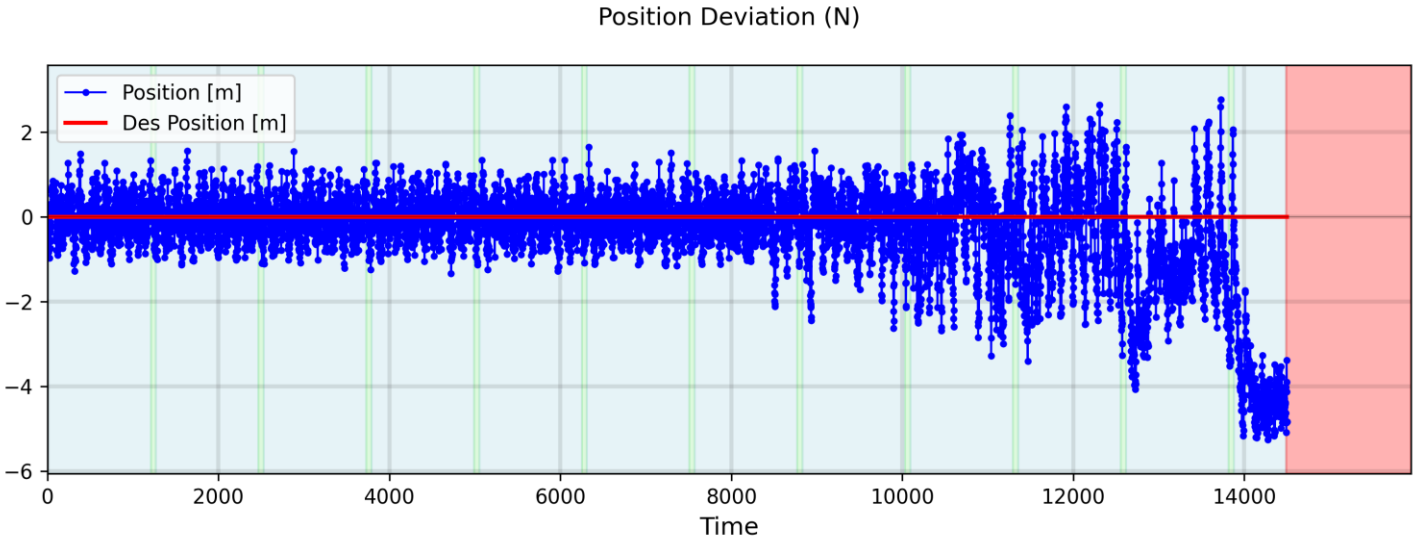
4.1.1 Bow sector



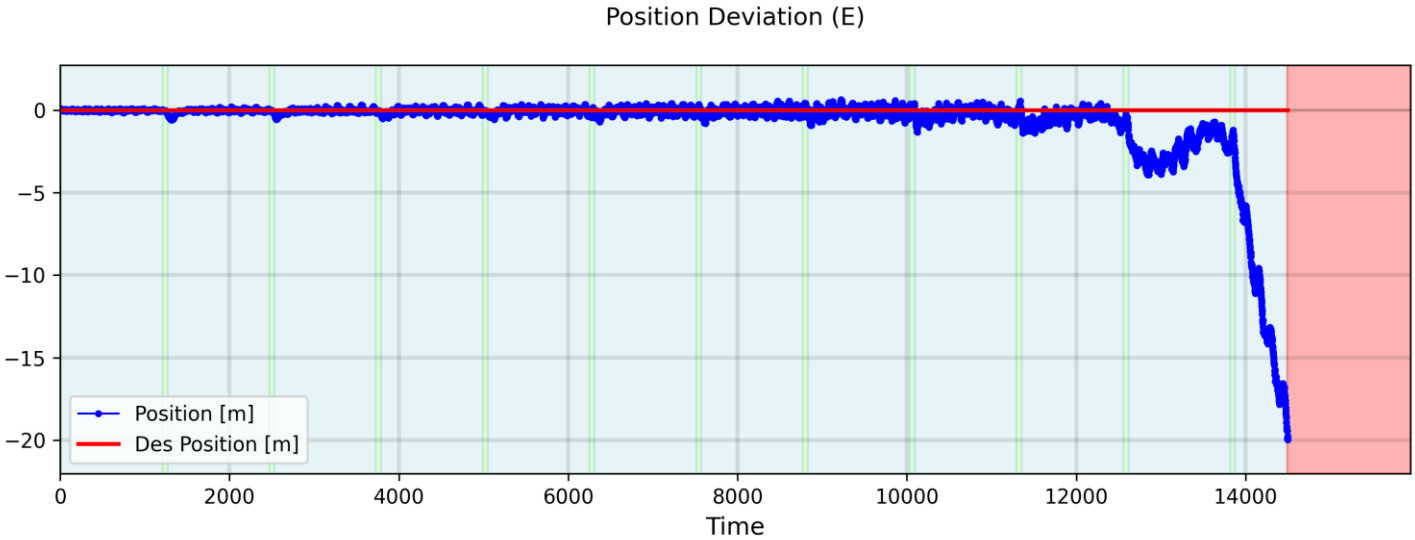
Picture 1: Disturbing force direction, where X axis - time in seconds, Y axis – direction in degrees



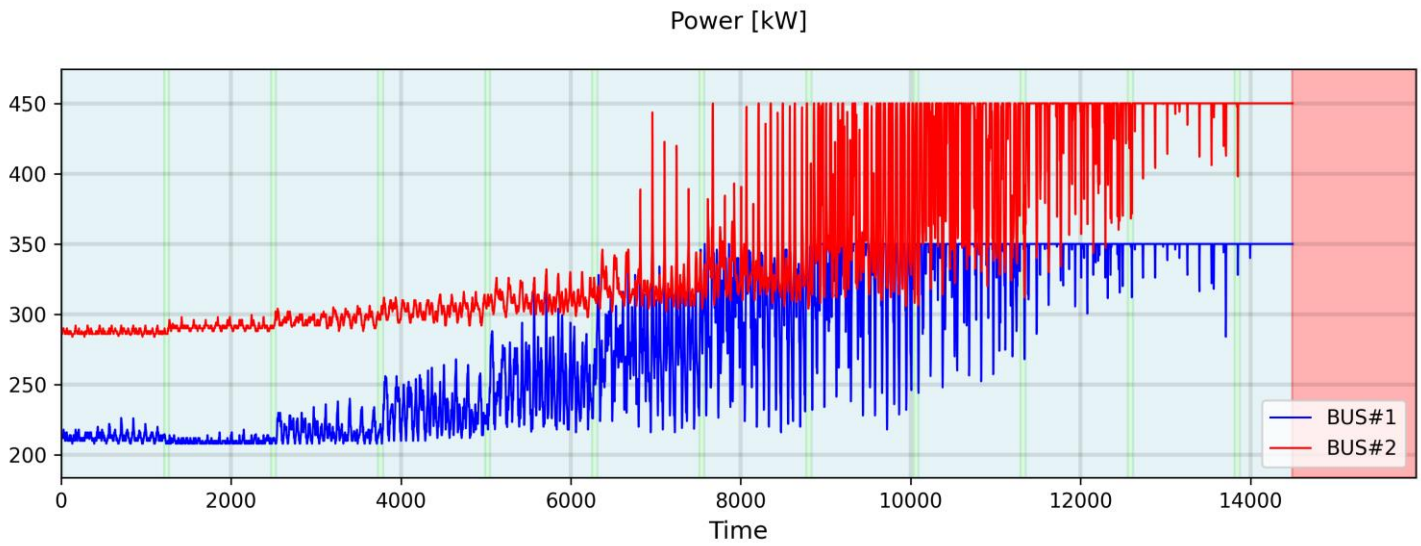
Picture 2: Heading deviation, where X axis – time in seconds, Y axis – heading in degrees
("+" – deviation to the right, "-"– deviation to the left side), Red line shows the set heading = 0˚



Picture 3: Longitudinal position deviation, where X axis – time in seconds, Y axis – deviation in meters



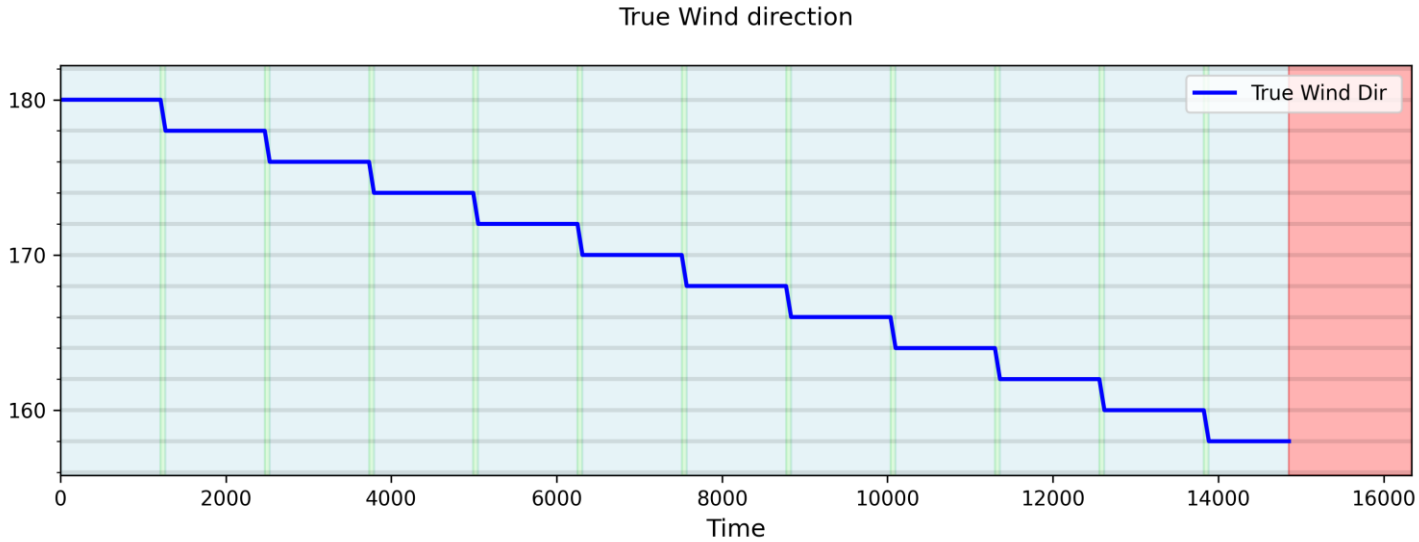
Picture 4: Transverse position deviation, where X axis – time in seconds, Y axis – deviation in meters



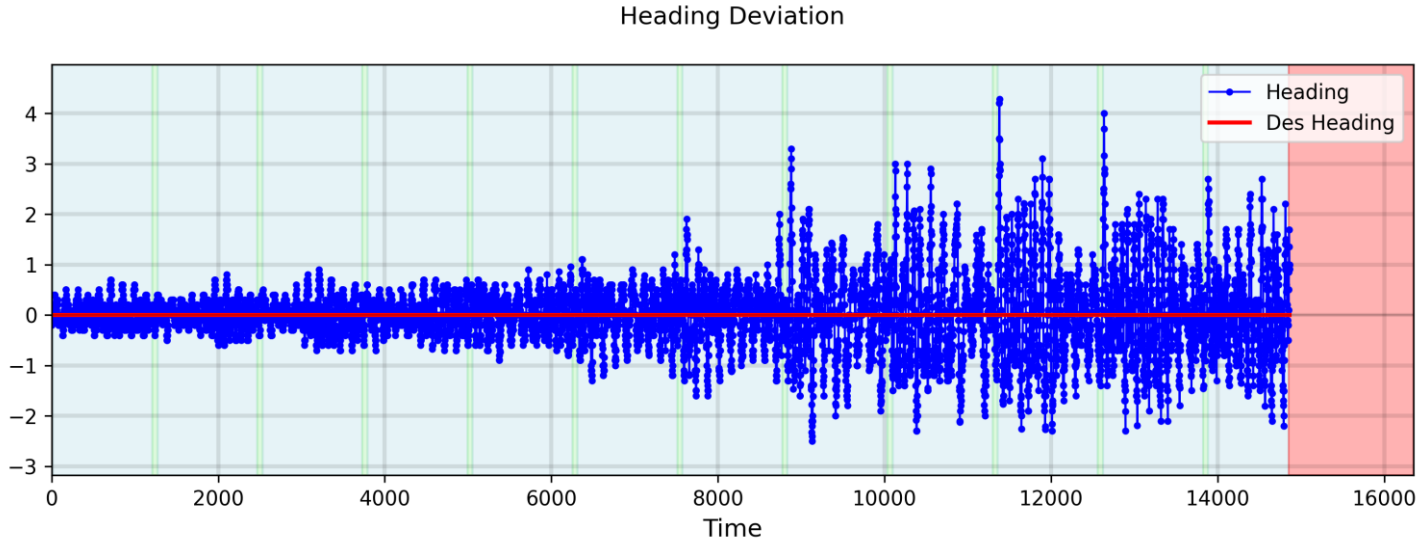
Picture 5: Propulsion load in power buses, where X axis – time in seconds, Y axis – power in kW

Conclusion: Based on the test results, the vessel is capable of maintaining its position when the environmental disturbing force is applied within a ±20° sector relative to the bow. Within this operational sector, the maximum observed heading deviation is 1.7°, while the maximum position deviations are within 5.1 meters radius.

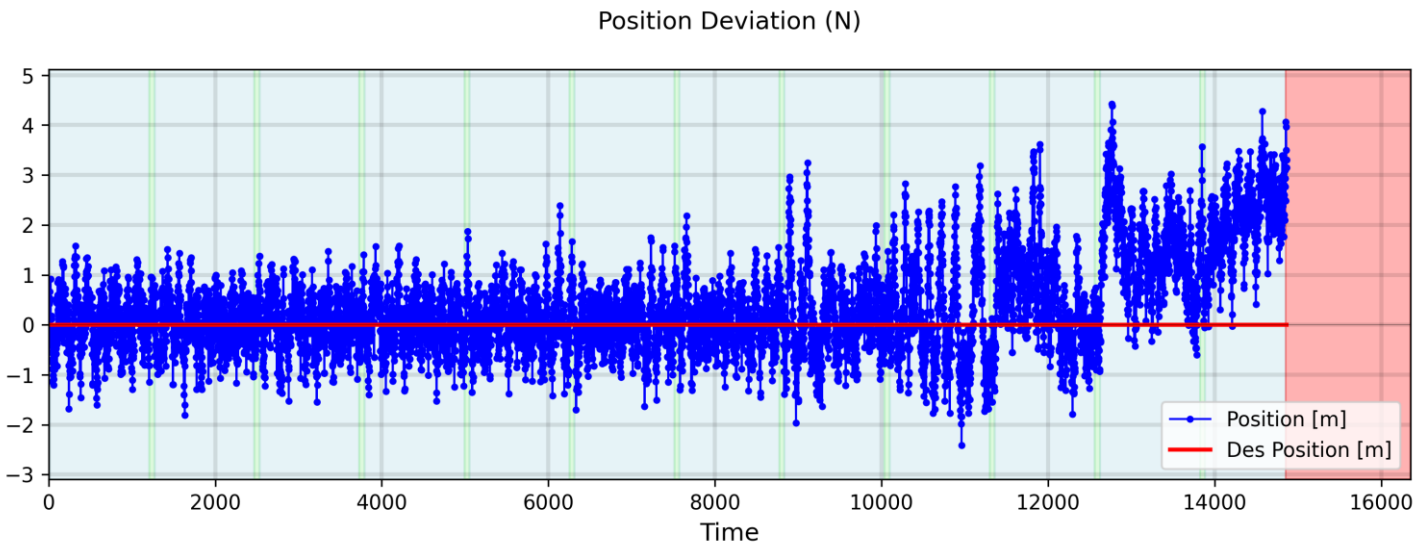
4.1.2 Stern sector



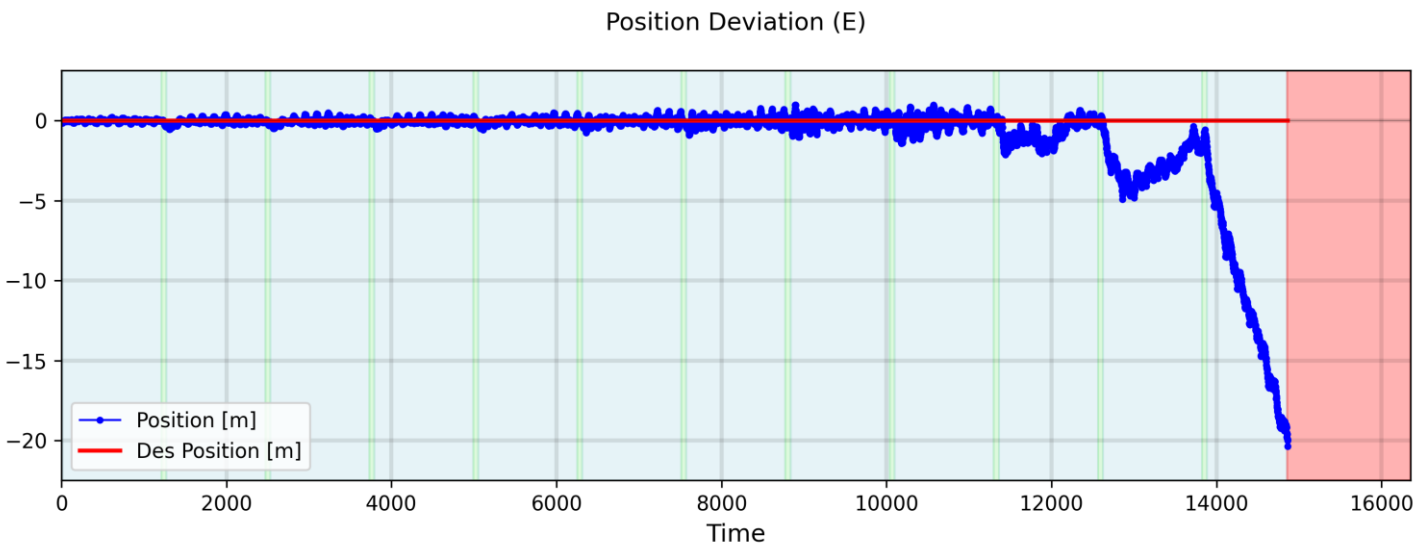
Picture 6: Disturbing force direction, where X axis - time in seconds, Y axis – direction in degrees



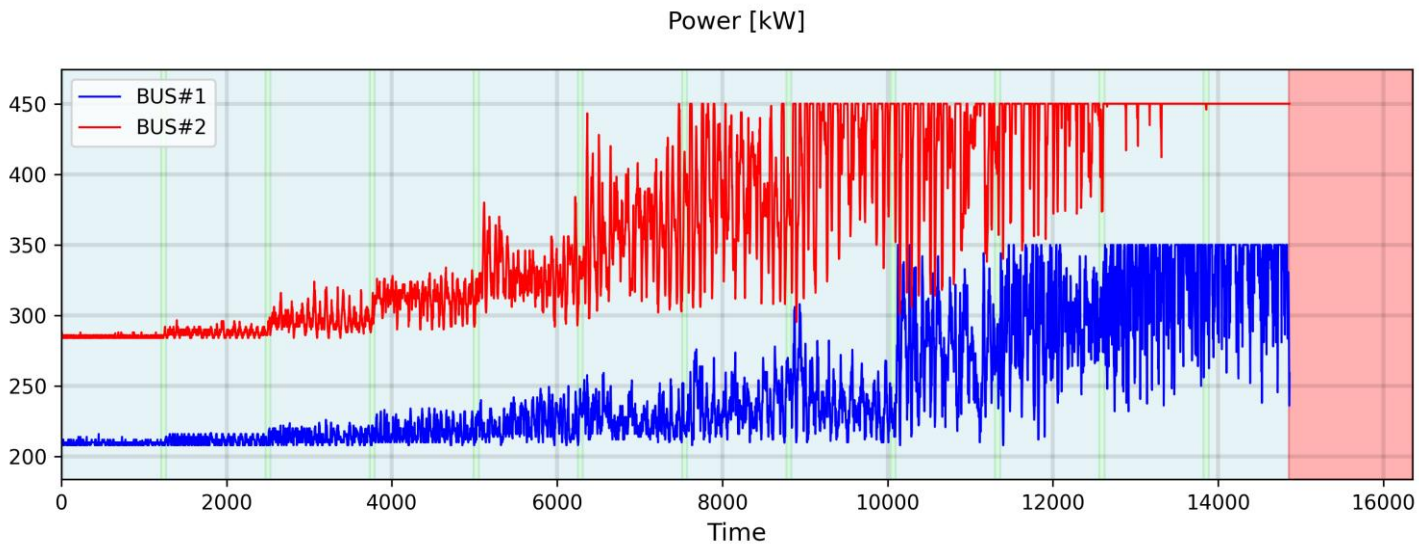
Picture 7: Heading deviation, where X axis – time in seconds, Y axis – heading in degrees
(“+” – deviation to the right, “-“– deviation to the left side), Red line shows the set heading = 0˚



Picture 8: Longitudinal position deviation, where X axis – time in seconds, Y axis – deviation in meters



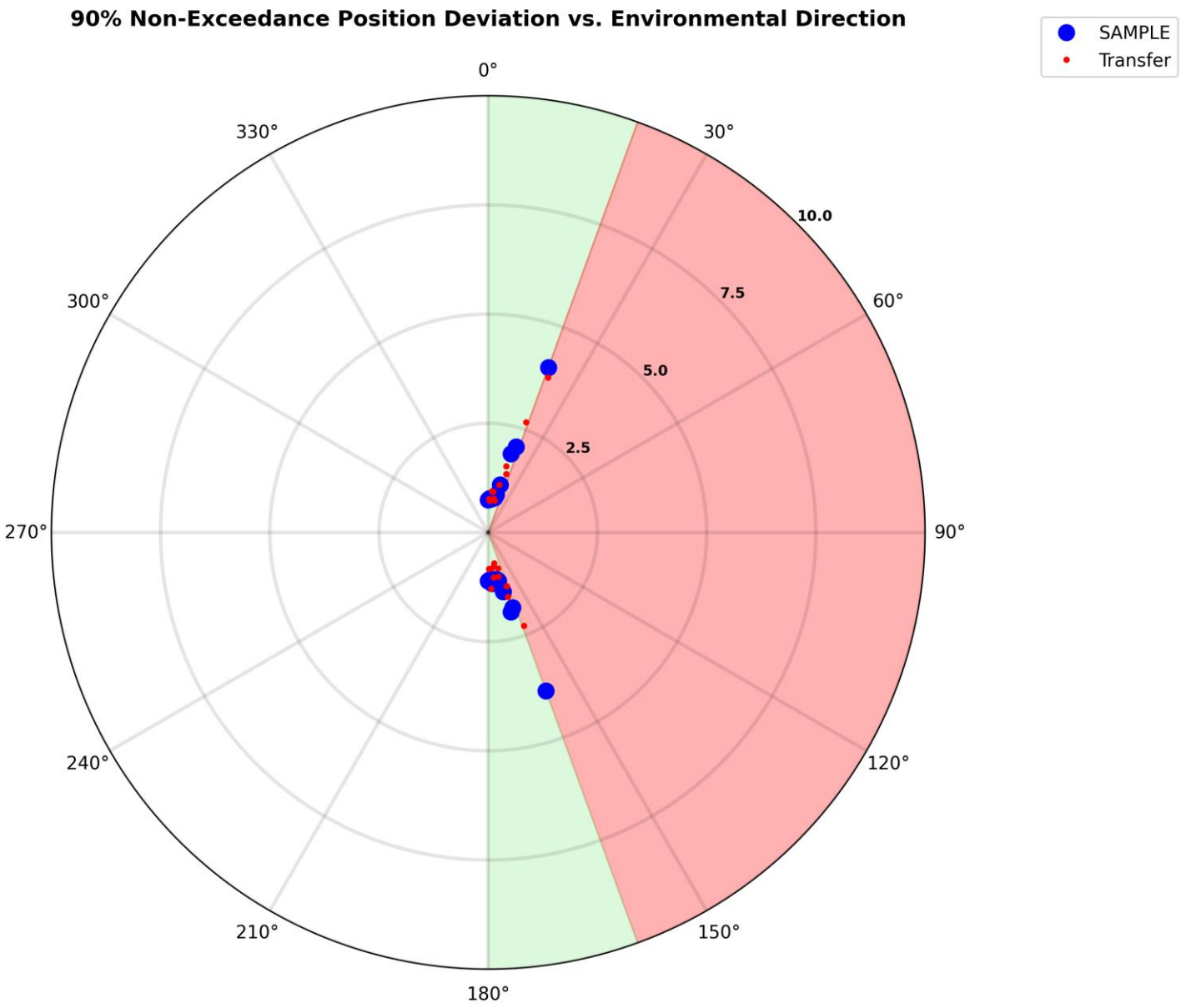
Picture 9: Transverse position deviation, where X axis – time in seconds, Y axis – deviation in meters



Picture 10: Propulsion load in power buses, where X axis – time in seconds, Y axis – power in kW

Conclusion: Based on the test results, the vessel is capable of maintaining its position when the environmental disturbing force is applied within a ±20° sector relative to the stern. Within this operational sector, the maximum observed heading deviation is 4.3°, while the maximum position deviations are within 5.4 m radius.

4.1.3 General result for Test mode #1



Picture 11: 90% Non-Exceedance Position Deviation vs. Environmental Direction. 90th percentile position deviation for each tested direction. Blue dots: stable environmental conditions (20-min samples). Red dots: during direction transitions. Green sectors indicate successful station-keeping; red sectors indicate capability limits exceeded (position deviation > 20 m threshold).

As a result of the tests conducted using the Fixed Heading mode under the weather conditions and disturbing force parameters described in 3.1, the vessel maintained its position within a radius of 5.4 meters. More detailed performance values are provided in Table 1 below.

Table 1:

Sample #	Wind & Current direction (deg)	Continuous test interval (min)	Max. Ship position deviation (m)	Max. Ship heading deviation (deg)
1	0.00	20	1.5	0.7
2	2.00	20	1.6	0.8
3	4.00	20	1.6	0.9
4	6.00	20	1.3	0.9
5	8.00	20	1.4	1.1
6	10.00	20	1.7	1.2
7	12.00	20	2.1	1.3
8	14.00	20	2.6	1.5
9	16.00	20	3.3	1.5
10	18.00	20	3.5	1.6
11	20.00	20	5.1	1.7
12	22.00	20	20+ (Failed)	1.8
13	158.00	20	20+ (Failed)	2.8
14	160.00	20	5.4	3.9
15	162.00	20	2.8	4.3
16	164.00	20	2.8	3.0
17	166.00	20	2.9	3.3
18	168.00	20	1.9	2.0
19	170.00	20	2.2	1.3
20	172.00	20	1.8	1.0
21	174.00	20	1.9	0.8
22	176.00	20	1.9	0.8
23	178.00	20	1.7	0.7
24	180.00	20	2.2	0.7

Conclusion: Based on the simulation and diagram outputs, the KCG 3000-ton class vessel (Pacific No.20, 3020) can maintain position and heading under Sea State 5 environmental conditions (mathematically simulated) within the following effective heading sectors:

- Bow sector: ±20° (approximately 0°–20°)
- Stern sector: ±20° (approximately 160°–200°)

Performance summary

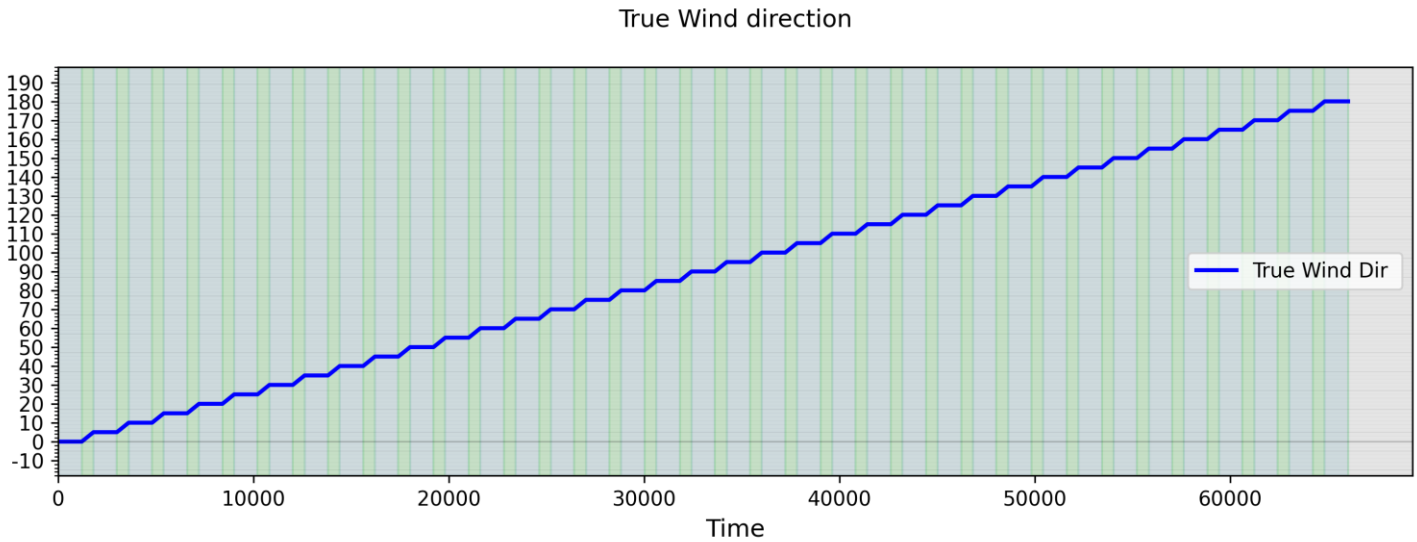
- Maximum position deviation: radius 5.40 m
- Maximum heading deviation: ± 4.3°

5 Test mode #2

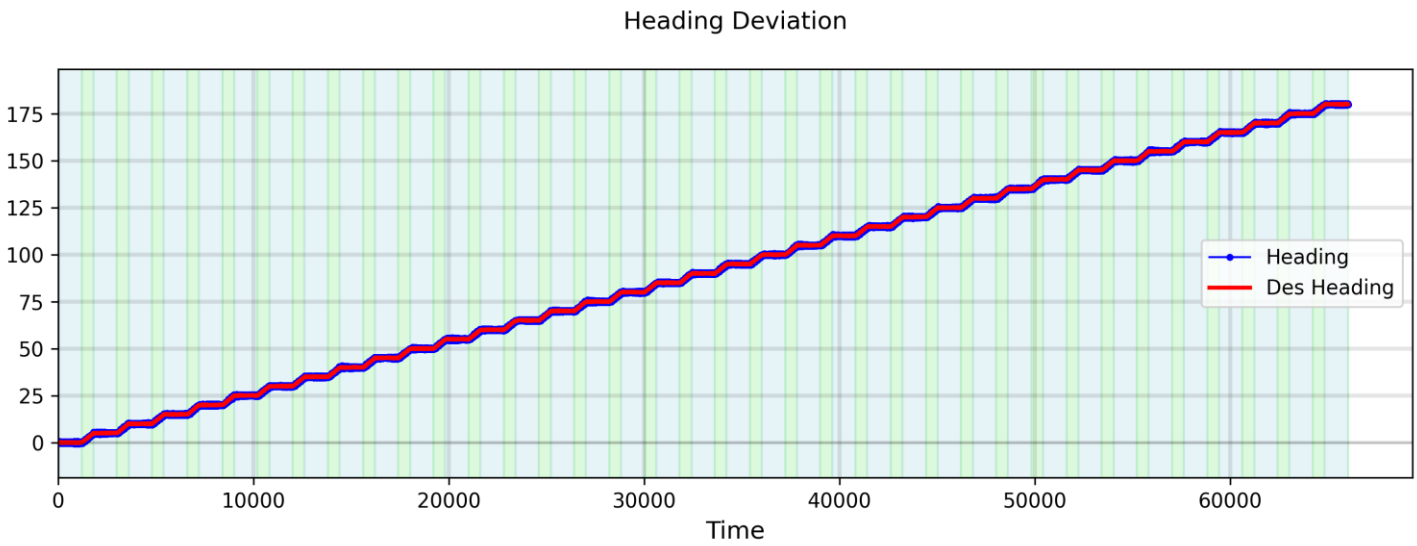
The objective of this test is to evaluate the position-keeping accuracy of the dynamic positioning system when operating in optimal-heading mode under varying environmental disturbance directions. The disturbance force direction is systematically varied over a range of 0° to 180°, and the resulting position-keeping performance of the vessel is assessed.

- vessel is set in conditions according to item 3.1 and 3.2 (Test mode #2).
- DP mode is active, and the vessel keeps the heading and position.
- initial time value for logging is set in 0 seconds.
- disturbing force change speed is 0.5deg/sec
- direction of the current is always collinear with the wind direction and they change simultaneously
- test interval is 5 degrees, the test duration is 20min
- test becomes failed if the vessel moves from the initial point more than 15m, after that the logging stops automatically
- test is conducted in a 0-180˚ range of disturbing force directions, because it is supposed that the results in a 180-360 ˚ part are identical due to the symmetric configuration and can be mirrored.

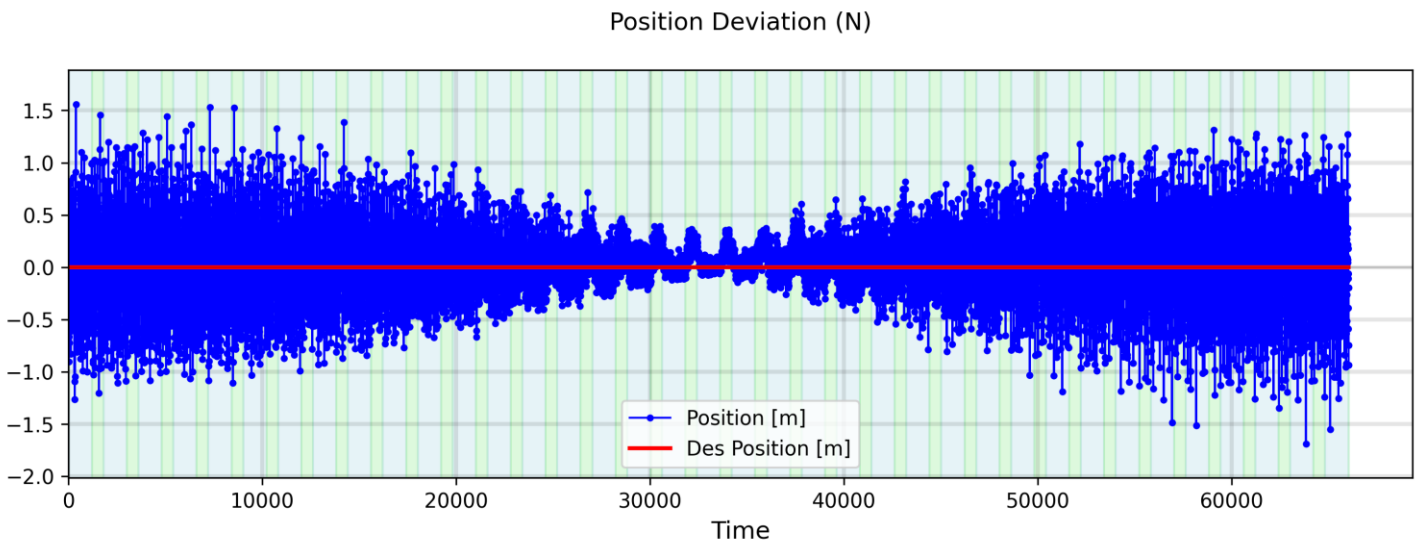
5.1 Test mode #2 results



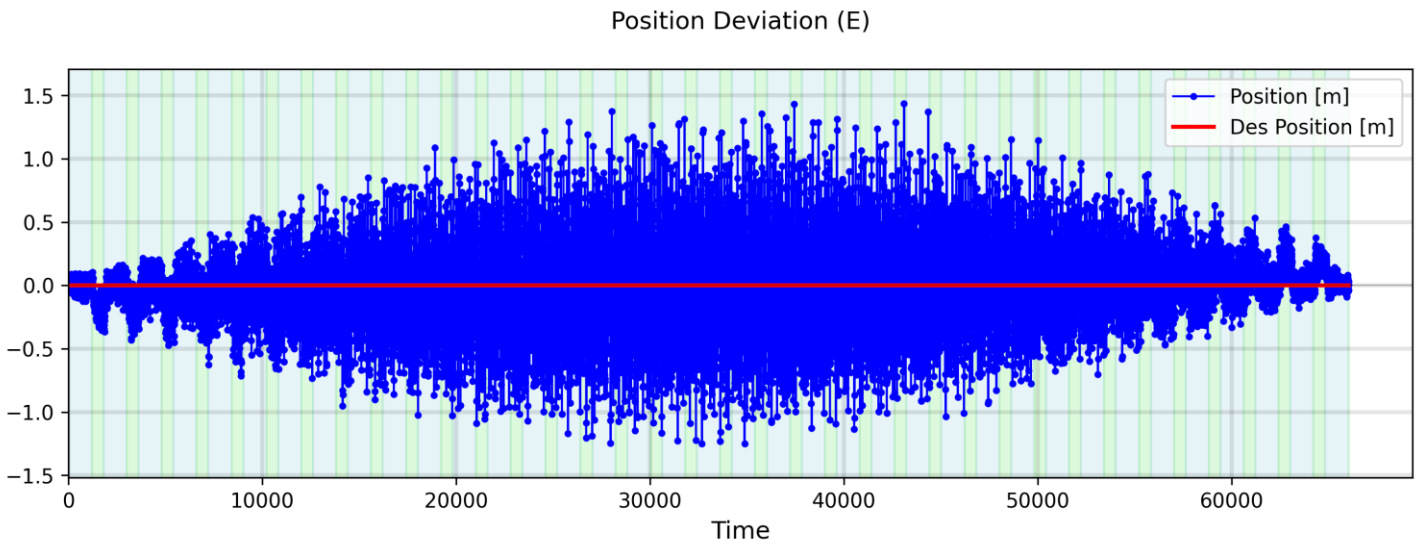
Picture 12: Disturbing force direction, where X axis - time in seconds, Y axis – direction in degrees



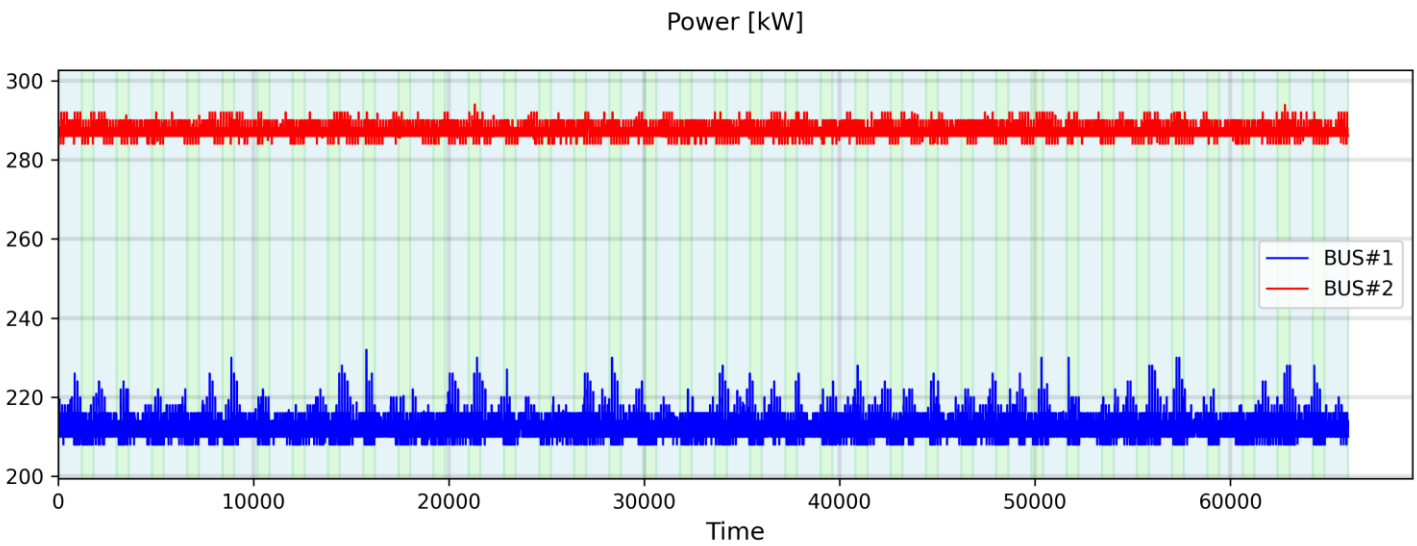
Picture 13: Heading deviation, where X axis – time in seconds, Y axis – heading in degrees.
Red line shows the set heading = optimal heading in each time period



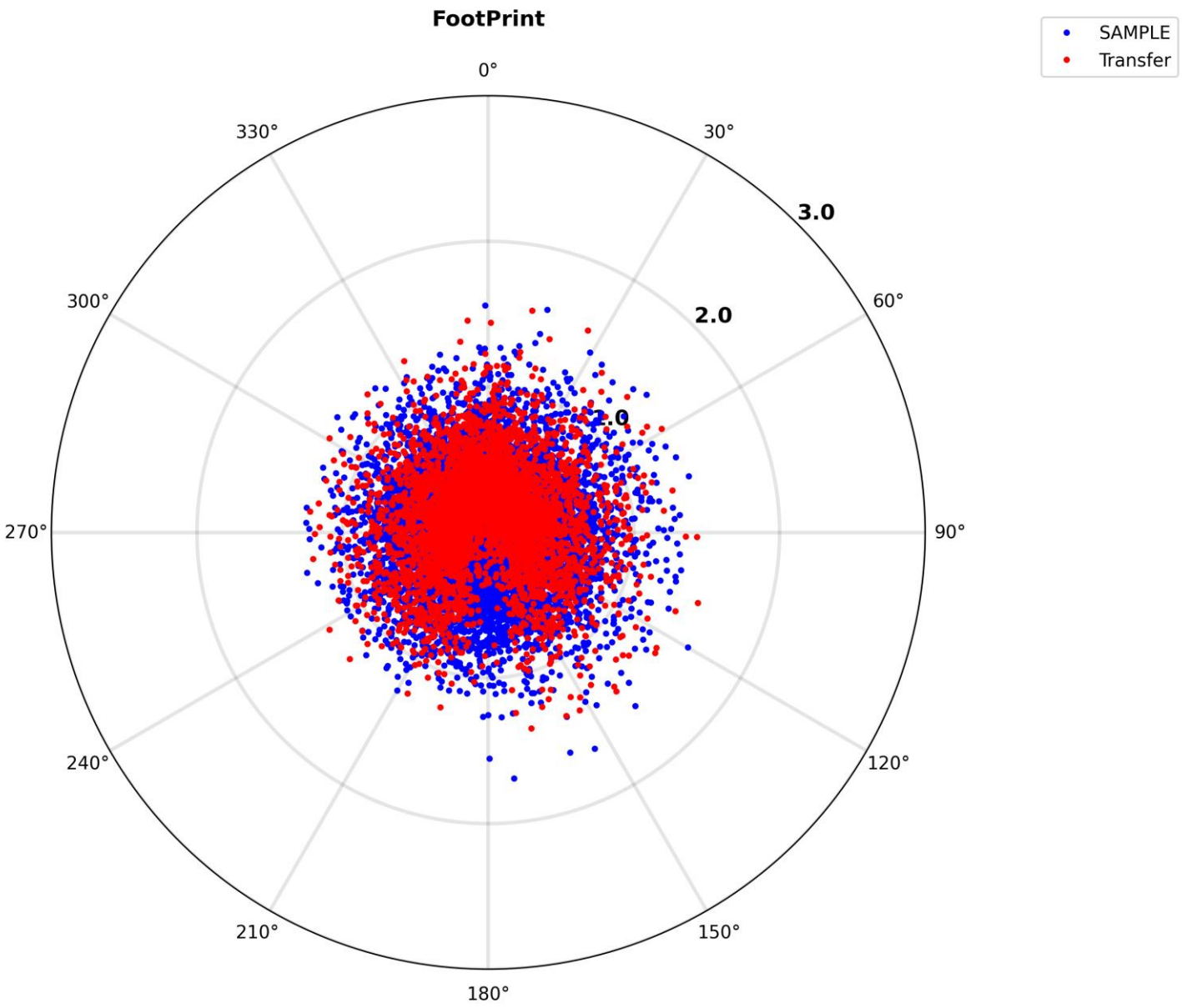
Picture 14: Longitudinal position deviation, where X axis – time in seconds, Y axis – deviation in meters,
Red line shows the set position point X=0



Picture 15: Transverse position deviation, where X axis – time in seconds, Y axis – deviation in meters
Red line shows the set position point Y=0



Picture 16: Propulsion load in power buses, where X axis – time in seconds, Y axis – power in kW



Picture 17: Footprint of the vessel position during all the test cases according to the disturbing force direction, where blue dots – vessel coordinates when the disturbing force is fixed, red dots – vessel coordinates when the disturbing force changes

5.2 General test result in Test mode #2

As a result of the tests conducted using the Optimal Heading mode under the weather conditions and disturbing force parameters described in in 3.1, the vessel maintained its position within a radius of 1.7 m. More detailed performance values are provided in Table 1 below.

Table 2:

Sample #	Wind & Current direction (deg)	Continuous test interval (min)	Max. Ship position deviation (m)	Max. Ship heading deviation (deg)
1	**0.0**	**20**	**1.56**	**0.86**
2	**5.0**	**20**	**1.20**	**0.71**
3	**10.0**	**20**	**1.30**	**0.53**
4	**15.0**	**20**	**1.41**	**0.55**
5	**20.0**	**20**	**1.58**	**0.75**
6	**25.0**	**20**	**1.22**	**0.79**
7	**30.0**	**20**	**1.42**	**0.33**
8	**35.0**	**20**	**1.39**	**0.66**
9	**40.0**	**20**	**1.34**	**0.91**
10	**45.0**	**20**	**1.23**	**0.65**
11	**50.0**	**20**	**1.44**	**0.77**
12	**55.0**	**20**	**1.40**	**0.83**
13	**60.0**	**20**	**1.28**	**0.74**
14	**65.0**	**20**	**1.35**	**0.46**
15	**70.0**	**20**	**1.39**	**0.49**
16	**75.0**	**20**	**1.43**	**1.08**
17	**80.0**	**20**	**1.20**	**0.72**
18	**85.0**	**20**	**1.32**	**0.63**
19	**90.0**	**20**	**1.25**	**0.89**
20	**95.0**	**20**	**1.30**	**0.90**
21	**100.0**	**20**	**1.35**	**0.79**
22	**105.0**	**20**	**1.33**	**0.89**
23	**110.0**	**20**	**1.32**	**0.94**
24	**115.0**	**20**	**1.34**	**0.87**

25	120.0	20	1.58	0.82
26	125.0	20	1.35	0.67
27	130.0	20	1.28	0.88
28	135.0	20	1.41	0.87
29	140.0	20	1.56	0.84
30	145.0	20	1.33	0.52
31	150.0	20	1.40	0.88
32	155.0	20	1.65	0.97
33	160.0	20	1.61	0.69
34	165.0	20	1.27	0.51
35	170.0	20	1.28	0.87
36	175.0	20	1.70	0.88
37	180.0	20	1.55	0.80

6 Comparison: Test Mode #1 vs. Test Mode #2

Test Mode #1 (Fixed Heading) represents classical DP operation where both position and heading are set by the operator. The vessel maintains a constant heading (0° in this test) regardless of environmental direction.
Test Mode #2 (Optimal Heading) allows the DP system to automatically adjust heading to minimize environmental loads by orienting the vessel's bow toward the resultant environmental force, while maintaining the commanded position.
Performance Summary:

Parameter	Fixed Heading	Optimal Heading	Improvement
Directional Coverage	±20° bow/stern only	Full 360°	-
Max Position Deviation	5.1 m (at 20°)	1.7 m (at 175°)	67% reduction
Max Heading Deviation	1.7°	1.1°	35% reduction

Operational Considerations - Environmental Change Rate Dependency:
The Optimal Heading mode performance shown in this study is based on the specific test parameters: environmental direction changes of 5° per step with stable conditions maintained for 20 minutes between steps.
Important limitation: In real operational scenarios with faster environmental direction changes or larger angular steps, performance may degrade significantly. During rapid environmental shifts, the vessel must reorient to track the changing optimal heading while simultaneously maintaining position. If the rate of environmental change exceeds the vessel's heading response capability, position excursions will increase substantially and may result in loss of position (exceeding the 20 m acceptance threshold).
Scenarios requiring further evaluation:
- Rapid weather front passages (environmental direction change > 15-30°/min)
- Larger environmental direction steps (> 15-30° increments)
- Combination of rapid direction change and increasing environmental intensity

Recommendations:
- Optimal Heading mode: Preferred for general station-keeping when heading is not operationally constrained. Provides superior performance under gradually varying or stable environmental conditions.
- Fixed Heading mode: Required when specific vessel orientation is necessary (cargo operations, proximity to structures). Limit operations to bow/stern sectors (±20°) under Sea State 5 conditions.
- Further testing recommended: If operations are planned in areas with rapid environmental changes (e.g., squall lines, frontal systems), additional simulations with higher direction change rates should be conducted to establish safe operational limits.